Multitasking Between Realities: An Examination of Multitasking Using Augmented Reality Under Different Task Demand Levels

Peiyu Zhang[1] and Sara Lu Riggs[1]

[1]*Department of Systems and Information Engineering, University of Virginia, Charlottesville, VA, USA*

This study investigated how optical see-through head-mounted displays (OST-HMDs) affect multitasking performance and perceived workload when switching between real-world and augmented reality (AR) contexts. Using a multitasking simulation on the Magic Leap 2 OST-HMD, participants performed a series of monitoring and input tasks under varying levels of task demand. Results showed that AR use led to significantly higher error rates, longer response times, and greater perceived workload compared to a desktop-only condition. Multitasking with AR also amplified the impact of increased task demand on workload and error rate in some tasks. These findings underscore the need for more careful design of AR systems, particularly in high-stakes environments where frequent context switching and multitasking are required.

INTRODUCTION

The adoption of virtual interfaces for presenting information and enriching interaction has been increasingly popular in recent years. Augmented Reality (AR) brings the benefits of virtual interfaces and simulated content into the real world by superimposing them onto real surfaces and spaces. AR has a wide range of applications, including placing virtual instructions near hands and tools to guide maintenance workers (Palmarini et al., 2018), overlaying information onto patients to assist surgical procedures (Kim et al., 2017), and integrating with traditional 2D displays to extend collaborative environments and present 3D views (Fröhler et al., 2022). One of the most widely used methods for displaying AR content is through optical see-through head-mounted displays (OST-HMDs) like the Microsoft HoloLens. These devices allow users to view the real environment directly, overlaid with stereoscopic virtual images. While this approach may enable more natural interaction and reduce motion sickness compared to video see-through displays, it also introduces challenges. Given the real and virtual views exist in separate displays and layers, continuous attention switching between virtual and real contexts may increase perceptual and cognitive workload (Gabbard et al., 2018).

To design AR displays that effectively support real-world applications—such as AR-assisted surgery—it is important to understand how OST-HMDs influence multitasking performance and perceived workload, particularly in environments that require frequent task switching. This study examines the effects of OST-HMDs on multitasking performance in mixed-reality environments, focusing on two key research questions: (a) What is the effect of OST-HMDs on multitasking when switching between realities (physical world and AR)? (b) What is the effect of OST-HMDs on multitasking under low and high levels of task demand?

BACKGROUND

Previous research has investigated the impact of context switching in AR environments, particularly with monocular OST-HMDs. Studies have reported decreased performance on visual search tasks immediately after introducing a virtual interface (Huckauf et al., 2010) or when significant spatial separations exist between virtual and real objects (Gabbard et al., 2018). These findings suggest that integrating virtual content into real-world tasks can introduce new challenges. However, fewer studies have examined the effects of frequent task switching on both performance and perceived workload when using modern binocular OST-HMDs. Recent advancements in HMD technology have improved image quality and depth perception, potentially mitigating some of the challenges observed in prior studies. However, understanding how these newer devices influence multitasking performance is crucial for designing AR interfaces that support high-stakes applications such as surgery, manufacturing, and emergency response. This is the goal of this study.

METHODS

The Magic Leap 2 OST-HMD with 1440 x 1760 pixels per eye resolution and a field of view of 70° diagonal was used to display the AR interfaces. A desktop display was used to provide real-world

information. A spacecraft flight testbed inspired by the NASA MATB-II was developed and shown in Figure 2. The bottom three tasks required the user to notice any deviation or command (i.e., buttons toggling from on to off, bars going above a certain threshold, pressure indicators entering the dangerous red zone, and entering requested parameters for setting) and respond using a mouse. The top target detection task required the user to select cannons with a mini keyboard and shoot incoming red targets. There were two blocks: (1) desktop only (all tasks were shown on the desktop) and (2) desktop and AR (the bottom three tasks were shown in AR). At any given point in time, the top target detection task and one of the bottom three tasks would require a response. The input for the testbed was generated randomly, but the bottom three tasks all occurred the same number of times. Each block had two scenarios, one low task demand and one high task demand, achieved by varying the length of allowed response time and the number of responses needed. Each scenario lasted three minutes and after each scenario, participants completed a NASA-TLX questionnaire to assess workload. At the conclusion of the study, participants completed a debriefing questionnaire and were compensated for their time.

Data from 19 participants were collected. A 2 × 2 repeated measures ANOVA was conducted to examine the effects of condition (desktop, AR) and task demand (low, high) on overall perceived workload, as well as the mean response time and error rate for each task. Sphericity was assessed, and if violated, a Greenhouse-Geisser correction was applied. Post-hoc tests were performed for significant interaction between condition and task demand.

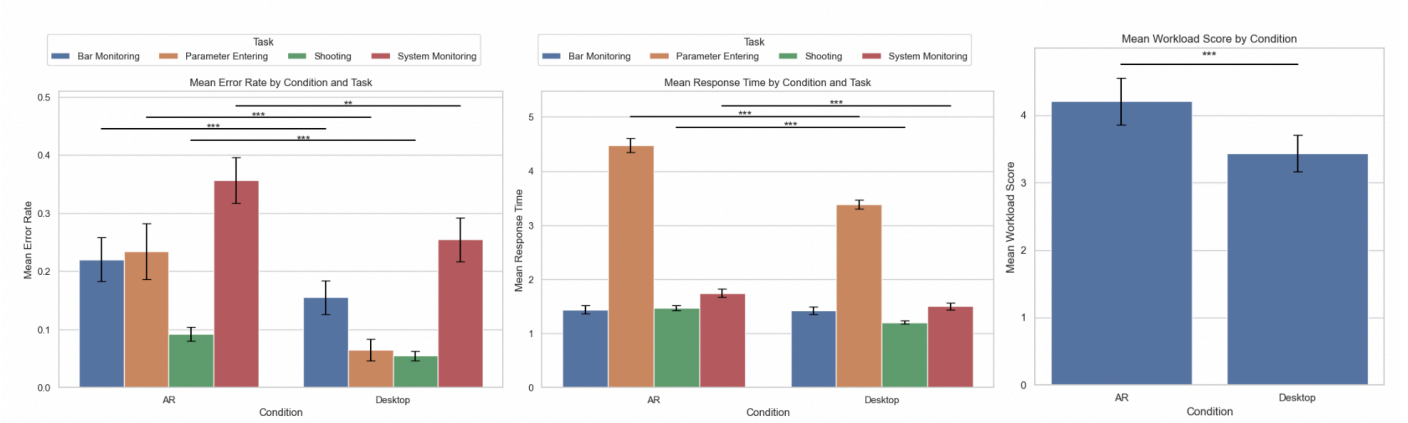

Figure 1: Bar plots showing mean error rates, response times, and workload scores (± standard errors) as a function of layout condition Lines between bars indicate significant pairwise comparisons. Significance bars are shown only for statistically significant comparisons. * indicates $p < .05$, ** indicates $p < .01$, and *** indicates $p < .001$.

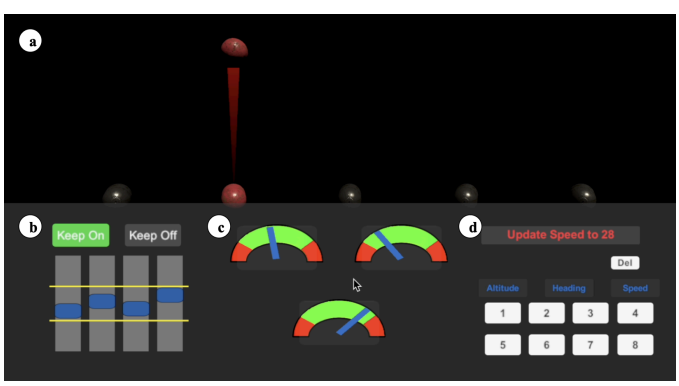

Figure 2: The spacecraft flight testbed with four subtasks: (a) Shooting (target detection) task, requiring users to select the aligned cannon and shoot a laser; (b) System monitoring task, requiring users to detect abnormalities in button toggling and bar floatation; (c) Bar monitoring task, requiring users to detect when indicators enter the red zone; (d) Parameter entering task, requiring users to input requested parameters.

OUTCOME

Figure 1 illustrated the effect of using an OST-HMD, which required switching between augmented reality and real-world contexts, on multitasking performance and perceived workload. Error rates across all four tasks were significantly higher in AR and high task demand conditions. A significant interaction was observed between task demand and condition for the shooting, bar monitoring, and

parameter entering tasks. In these tasks, error rates increased more sharply under AR under higher task demands compared to the desktop condition, although both conditions showed significant increases.

Response times were significantly lower in the desktop condition and under high task demands for shooting and system monitoring tasks. For bar monitoring, only high task demand led to a significant reduction in response time, while for parameter entering, only the desktop condition resulted in significantly faster responses. In the shooting task, high task demand in the desktop condition reduced response time less than in AR because the shooting response time in the low-task-demand desktop condition was relatively fast. For parameter entering, the effect of task demand on response time was significant only in the desktop condition, where high task demand led to slower responses. This was likely due to the parameter entering task being more challenging than the other tasks. Finally, workload scores, measured using the NASA-TLX 10-point scale, were significantly higher in AR and under high task demand. The increase in perceived workload due to higher task demand was more pronounced in AR than in the desktop condition, though both showed significant effects.

CONCLUSION

The results of this study found that switching between AR and real-world contexts leads to significant performance decrements, including higher error rates across all tasks and longer response times in three out of four tasks. Additionally, perceived workload was also higher in AR. Increased task demands generally resulted in higher error rates and greater workload, though it also led to shorter response times. Performing tasks in AR further amplified the effects of task demand on workload and error rate for some tasks. These findings suggest that designers should carefully consider the additional cognitive burden and performance impairments when requiring users to multitask across AR and real-world environments, especially under high task demands. Moreover, the study showed that performance and workload were significantly and adversely affected even when display locations remained unchanged after switching from physical to virtual display in AR.

Overall, the current study design only included one virtual and one real-world interface, and both interfaces were stable and presented on the same plane. Future research may consider exploring beyond these constraints and investigating alternative user input methods beyond the traditional mouse and keyboard to gain insights into how performance and workload could be further impacted.

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